HYBRID PARTICIPATION INNOVATION LAB · TOR VERGATA · FULL PROJECT

Fair Share: an AI energy assistant designed with its tenants

A hybrid participatory process for a solar energy community in Tor Bella Monaca, Rome

Next Wave

Mahsa Shahi · Kuhyar Saeedi · Farida Safdari · Gelila Tsehaye Melesse · Md Nazmul Islam · Pranto Argha Sarker

AGENDA

Ten steps, from problem to evaluation

01	Define the innovation problem	14 Oct
02	Map the stakeholder ecosystem	14 Oct
03	Co-define the participatory challenge	28 Oct
04	Decide the level of participation	28 Oct
05	Select methods by purpose	4 Nov
06	Design the hybrid architecture	4 Nov
07	Citizen science component	18 Nov
08	Prototype: the Fair Share Simulator	24 Nov
09	Facilitation and inclusion plan	2 Dec
10	Evaluation framework	9 Dec

01

Define the innovation problem

What the AI assistant is, who builds it, where it lands, what it promises and risks

QUESTION 1 OF 6

What is the technology, and what problem does it solve?

THE PROBLEM

Families in public housing struggle to pay energy bills.

Solar panels on the roof help little: most energy is used in the evening, when there is no sun, and tenants have no say in how the money is shared.

OUR TECHNOLOGY: AN AI ENERGY ASSISTANT

1 **Predicts the sun:** how much energy the roof will make, hour by hour.

2 **Advises families:** the best time to run the washing machine or other appliances.

3 **Shares the money fairly,** using rules the tenants decide together.

QUESTION 2 OF 6

Who develops or promotes it?

A new idea

Nobody has built this yet. Our project designs it together with the people who will use it.

It builds on

Solar forecasting and smart plugs from energy and smart-home companies, and research from universities such as Tor Vergata.

Supporters and owners

The EU, the Italian state and GSE fund energy communities. The City of Rome supports them. ATER owns the buildings.

QUESTION 3 OF 6

Where will it be implemented?

A pilot in **one public housing complex** in Rome's eastern periphery, for example Tor Bella Monaca.

Large flat roofs

Plenty of space for solar panels that today produce nothing.

Families in need

Many low-income households who struggle with energy bills.

Near the university

Close to Tor Vergata, a possible research partner for on-site workshops.

QUESTION 4 OF 6

What benefits are expected?

Lower energy costs

More solar energy used at the right time means a bigger bonus for the community.

Fair sharing

The money is shared by rules the tenants themselves agree are fair.

Energy know-how

Families learn when and how to use energy better.

Cleaner energy

More local solar power, less CO2 and pollution.

QUESTION 5 OF 6

What uncertainties, risks or controversies may emerge?

Risk	In simple words
Fairness	People may disagree about what a "fair" share is.
Privacy	The AI needs private data about when people are at home.
Trust	Tenants may not trust a computer deciding about money.
Exclusion	Elderly people or people without smartphones may be left out.
Costs	It is not clear who pays for the devices and repairs.

QUESTION 6 OF 6

Why is participation necessary?

The AI decides who gets money, so the tenants must help write its rules.

What is "fair"?

Equal shares, more for poorer families, or more for those who shift their usage?

What data?

What the AI may know about their homes, and who can see it.

Advise or control?

Should it only suggest, or also switch appliances on?

02

Map the stakeholder ecosystem

Who is involved, who holds power, who is vulnerable, who is missing

1 · ACTORS

Who is involved?

Affected and users

Tenant families, elderly tenants, migrant families, day and night-shift workers, people with disabilities

Develop and build

AI tech provider, solar installer, Tor Vergata researchers

Own and manage

ATER Roma (owns the buildings), the CER association, Areti (Rome's grid operator)

Regulate

ARERA (energy regulator), GSE (incentive rules), Garante Privacy (data protection)

Fund and support

EU and Italian state, GSE incentive, City of Rome and Municipio VI, third-sector groups and the parish

May oppose or doubt

Distrustful tenants, tenant unions (e.g. SUNIA, Unione Inquilini), privacy advocates

2 · INTERESTS

What does each actor want, and fear?

Actor	Expects or gains	Fears or risks
Tenant families	Lower bills, a fair share of the money	Complex rules, hidden costs, unfairness
Elderly tenants	Simple help, no extra effort	Being left out by an app
ATER Roma	Lower common costs, a better image	Roof damage, new duties, conflicts
CER association	Members who join and stay	Low participation, admin burden
AI tech provider	A pilot, data, a product to sell	Low use, privacy fines, bad press
Researchers	Real data, impact, publications	Low trust, missing data, ethics issues
City of Rome	Less energy poverty, a visible success	Failure in a sensitive neighbourhood
Tenant unions	Protection of tenants' rights	Tenants paying or losing out

3 · KNOWLEDGE

Who knows what?

Scientific and technical

Researchers and the tech provider: forecasting, AI, solar systems

Institutional

GSE, ARERA, ATER, the City: rules, incentives, permits

Practical

Installer and Areti: roofs, the grid, meter data

Local and experiential

Tenants, committees, social workers: daily life in the buildings

Only tenants know what the AI can't measure:

their daily routines, who has no smartphone, which flats are empty during the day, and the promises that were broken before.

4 · POWER

Who controls what?

Control over	Who holds it	Where tenants stand
Resources	ATER owns the roofs, GSE pays the incentive	They own nothing in the system
Data	The tech provider and Areti	They produce it but don't control it
Decisions	ATER, the City, the CER board	They vote only if they join
Language	Experts: kWh, algorithms, incentives	Many can't follow the jargon
Access	Whoever has a smartphone, time and Italian	The most vulnerable have least

⚠ **Tenants have the least power, but the most at stake.**

5 · VULNERABILITY

Who could be harmed or left out?

Elderly living alone

Barrier: no smartphone, low digital skills

Risk: miss the benefits, lose control to automation

Low-income families

Barrier: no time, can't buy smart devices

Risk: carry costs, gain little

Migrant families

Barrier: language, distrust of institutions

Risk: not informed, not represented

Shift and day workers

Barrier: away when the sun shines, no time for meetings

Risk: the algorithm rewards people who stay home

People with disabilities

Barrier: inaccessible meetings and apps

Risk: automation touching devices they depend on

Tenants without a contract

Barrier: may not be allowed to join the CER

Risk: excluded completely

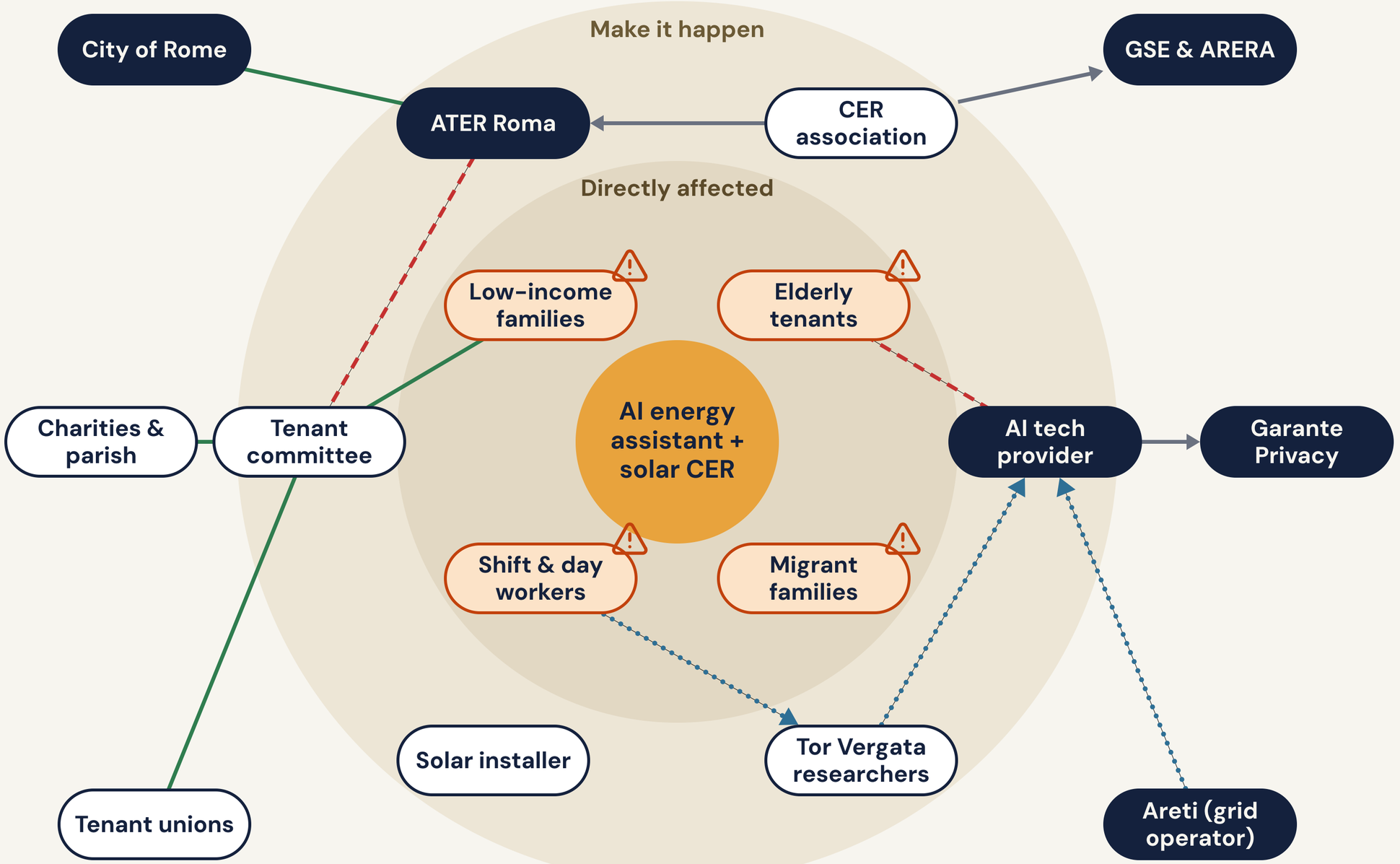
6 · MISSING VOICES

Who is usually absent, but should be involved?

Who	Why they are usually absent	Why they matter
Women carers	No time for evening meetings	They run most daily energy use at home
Young people	Never invited to housing decisions	Digital skills; they can help elders
Shift workers	Meetings clash with their hours	The fairness rule could penalise them
Migrants	Language and distrust	Different habits, needs and languages
Irregular tenants	No contract, afraid to speak	They live there and use energy too
Neighbours outside	Not members of the CER	They see the panels and may want to join

REQUIRED VISUAL

Stakeholder ecosystem map



How to read it

Rings: closer to the centre = more directly affected

- High influence
- Lower influence
- Vulnerable (warning sign)
- Alliance
- Depends on
- Conflict
- Knowledge or data flow

MAP NOTES

Why each actor should participate

Actor	Why they should take part
Tenant groups	They live with the system and must decide what "fair" means
Tenant committee	Bridges tenants and institutions, and builds trust
ATER Roma	Owns the roofs; must approve and maintain the system
CER association	Will run the system and apply the fairness rule
AI tech provider	Must build what tenants decide, not the other way round
Researchers	Bring methods and protect data quality and ethics
City of Rome	Can link the pilot to social services and scale it up
GSE & ARERA	Set the incentive rules the algorithm must respect
Garante Privacy	Defines which uses of home data are legal and safe
Areti	Provides meter data and the limits of the grid
Unions and charities	Reach missing voices and defend tenants' rights

WHAT THE MAP TELLS US

Three lessons for our process

1

Closest, but weakest

The people closest to the innovation have the least power over it. Our process must shift that balance.

2

Data is the conflict

The provider and the grid operator want data. Tenants fear losing privacy and control. Rules must be agreed together.

3

Trusted bridges

The committee, unions, charities and the parish can reach the missing voices that institutions can't.

Next, Step 3: rewrite the problem together with these actors.

03

Co-define the participatory challenge

The developers' problem is not the final problem

1 · THE ORIGINAL FORMULATION

How the developers would frame it

> **"How can an AI assistant get tenants to shift their energy use to sunny hours, so the community earns the highest incentive?"**

HIDDEN ASSUMPTIONS

- Tenants are flexible and at home in the day
- A bigger incentive means a better project
- An app is enough to reach everyone
- An equal split of the money is fair
- Sharing home data is unproblematic
- Tenants are users, not decision-makers

2 · WHAT OTHERS MIGHT ADD OR CONTEST

Imagined voices from our stakeholder map

ELDERLY TENANT

"I don't use apps. Who explains this to me, and can I say no?"

SHIFT WORKER

"I'm never home when the sun shines. Why should I get less?"

MIGRANT PARENT

"Will it be in my language? Who will see when we are at home?"

TENANT UNION

"Benefits must not depend only on behaviour. Need matters."

ATER ROMA

"Who maintains the roof and the devices, and who is liable?"

TECH PROVIDER

"Without good data the AI can't forecast. How much data is enough?"

3 · HOW THE PROBLEM SHIFTS

From optimising tenants to designing with them

From (developers' view)	To (shared view)
Maximise the incentive	Share benefits in a way tenants find fair
Change tenants' behaviour	Fit the system to different routines
Equal split by default	A fairness rule decided together
App for everyone	App, paper, phone and a person
Collect all useful data	A data charter tenants approve
Automate appliances	Advise by default, automate only by choice

4 · OUR SHARED PARTICIPATORY QUESTION

Under what conditions can an AI energy assistant help the different households of a public housing complex share solar energy and its benefits in a way they find fair, understandable and under their control?

5 · FIVE QUESTIONS WE ANSWER TOGETHER

What tenants will decide

- **Fairness:** how should the money be shared between households?
- **Data:** what can the AI know, who sees it, how long is it kept?
- **Control:** should the assistant advise, or also switch appliances?
- **Access:** how do people without a smartphone take part fully?
- **Governance:** who runs the community, and how can decisions be challenged?

04

Decide the level of participation

Different phases, different levels, and clear points where tenants decide

THE FIVE LEVELS, FOR OUR PROJECT

What each level means for tenants

1 · INFORM

How a CER and the AI work; tenants' rights

2 · CONSULT

Routines, needs, fears, what "fair" means to each household

3 · INVOLVE

Tenants set the agenda for what gets co-designed

4 · CO-DESIGN

Fairness rule, data charter, the assistant's messages and channels

5 · CO-CREATE

Knowledge from citizen science; governance of the community

LEVEL PER PHASE · PROPOSED 9-MONTH PROCESS

What tenants can change in each phase

Phase	Months	Level	What tenants can change
0 · Outreach	1	Inform	Raise questions that enter the agenda
1 · Listening	1–2	Consult	The list of problems and needs
2 · Open agenda	2	Involve	Which topics are co-designed, and in what order
3 · Fairness and data	3–4	Co-design	The fairness rule and the data charter
4 · Citizen science	3–6	Co-create	What the AI learns and how it forecasts
5 · Living lab	5–7	Co-design	Features, messages, paper and phone channels
6 · Governance	7–9	Co-create	Statutes, tenant seats, monitoring and appeals

DECISION POINTS

Four moments where tenants' choices bind

GATE 1 · MONTH 2

The agenda

Tenants vote on which issues are co-designed first.

GATE 2 · MONTH 4

Fairness rule

The tenant assembly adopts the rule; the provider must implement it.

GATE 3 · MONTH 4

Data charter

No household data is used beyond what the charter allows.

GATE 4 · MONTH 7

Go or no-go

After the pilot, tenants and ATER decide to scale, change or stop.

Our commitment: developers sign these gates in advance, so participation cannot be ignored afterwards.

WHY NOT CO-CREATE EVERYTHING?

Wide for everyone, deep for volunteers

Wide track

Every household can be informed, consulted and vote at the gates, with little time and no smartphone.

Deep track

A rotating group of volunteers co-designs and co-creates, with childcare, flexible times and recognition.

As Article 3 argues, not everyone has the time to co-create, but everyone should be able to decide at the gates.

05

Select methods by purpose

Every method does a job in the process, not decoration

THE METHOD CHAIN

Each method feeds the next

EXPLORE
Doorstep and walking interviews, focus groups

→

OPEN
Open Space day in the courtyard

→

DELIBERATE
Fairness assembly, fishbowl, scenarios

→

GENERATE
World Café and co-design workshops

→

KNOW
Citizen science: postcards, plugs, diaries

→

TEST
Living lab pilot, usability tests

We combine the methods REGACE used (Open Space, World Café and a fishbowl; Article 1), but in our own order, each where its job is needed.

EXPLORE · OPEN THE AGENDA

Listen first, then let tenants set the topics

Method	The task it performs	Who and where
Doorstep interviews	Reach people who never come to meetings	Every stairwell; with cultural mediators
Walking interviews	See roofs, meters and shared spaces through tenants' eyes	Tenants and researchers, on site
Focus groups by group	Hear each vulnerable group in a safe setting	Elderly, migrant parents, shift workers; at their hours
Participatory mapping	Locate sun, shade, broken things and gathering points	Paper map in courtyard plus online map
Open Space Technology	Tenants propose and choose the topics to co-design	One Saturday in the courtyard; streamed and summarised online

DELIBERATE · GENERATE IDEAS

Weigh the hard choices, then design together

Method	The task it performs	Who and where
Fairness assembly	Decide the fairness rule (Gate 2) with a group that mirrors the complex	About 25 tenants chosen by stratified lottery, 3 evenings
Scenario discussion	Compare rules on concrete families using our simulator	Inside the assembly, on a big screen and on paper
Fishbowl	Make disagreement visible and return results to all tenants	Open courtyard session after each assembly
World Café	Generate ideas for messages, alerts and the paper channel	Rotating tables, hosts per table
Co-design workshops	Turn ideas into screens, postcards and a data charter draft	Volunteers, designers, the provider

PRODUCE KNOWLEDGE · TEST AND ADAPT

Learn from real life, then try it for real

Method	The task it performs	Who and where
Energy postcards	Every household notes when it uses big appliances for two weeks	All households, paper or app
Smart plugs and diaries	Measure and explain the real use of key appliances	About 30 volunteer households
Data evenings	Tenants and researchers interpret the data together	Monthly, community room plus online summary
Living lab pilot	Run the assistant for 3 months and adapt it on the way	Volunteer households and the provider
Usability tests	Check that elderly and non-digital tenants can use it	One-to-one, at home, with a facilitator

WHY THESE METHODS

Three design choices

Not survey-only

A survey can't explain why a family cooks at 9 pm. Interviews, diaries and dialogue can.

Separate, then together

Vulnerable groups speak first in their own spaces, then meet the others with their points ready.

Lottery, not volunteers

For the binding fairness decision, a mirror of the complex beats the loudest residents.

06

Design the hybrid architecture

One process across the courtyard and the screen

FOUR DESIGN RULES

Digital for continuity, not to copy meetings

1 **Every online step has an offline twin**

Each vote, comment or survey also works on paper, by phone or in person.

2 **Meet to decide, go online to continue**

Decisions happen face to face; online keeps the thread between meetings.

3 **Information flows both ways**

Online comments are read aloud on site; on-site results are posted online within 48 hours.

4 **No smartphone needed**

Full participation is possible without any device, as Article 2 warns about digital capital.

PHASE BY PHASE

Where each phase happens

Phase	On-site (live)	Online (mostly async)	How information moves
Outreach	Doorstep visits, info stand	WhatsApp broadcast, short videos	Questions collected on both sides feed the agenda
Listening	Interviews, focus groups	Short survey, online map	Weekly summary on the noticeboard and online
Open agenda	Open Space day	Live stream, topic voting for 1 week	Online votes added to on-site results
Fairness and data	Assembly, fishbowl	Simulator, comments on drafts	Online comments read aloud at the next session
Citizen science	Data evenings	App diary, dashboard	Paper postcards digitised by ambassadors
Living lab	Home visits, tests	Assistant, feedback button	Monthly change log: "you said, we did"
Governance	CER assembly	Proxy and remote voting	Minutes in plain language, both channels

CHANNELS

Six channels, one conversation

Energy Corner

Noticeboard and suggestion box in the courtyard

Energy ambassadors

Young tenants who carry voices between paper and screen

Monthly bulletin

Printed, in several languages, under every door

WhatsApp groups

One broadcast list per language, voice notes welcome

Web platform

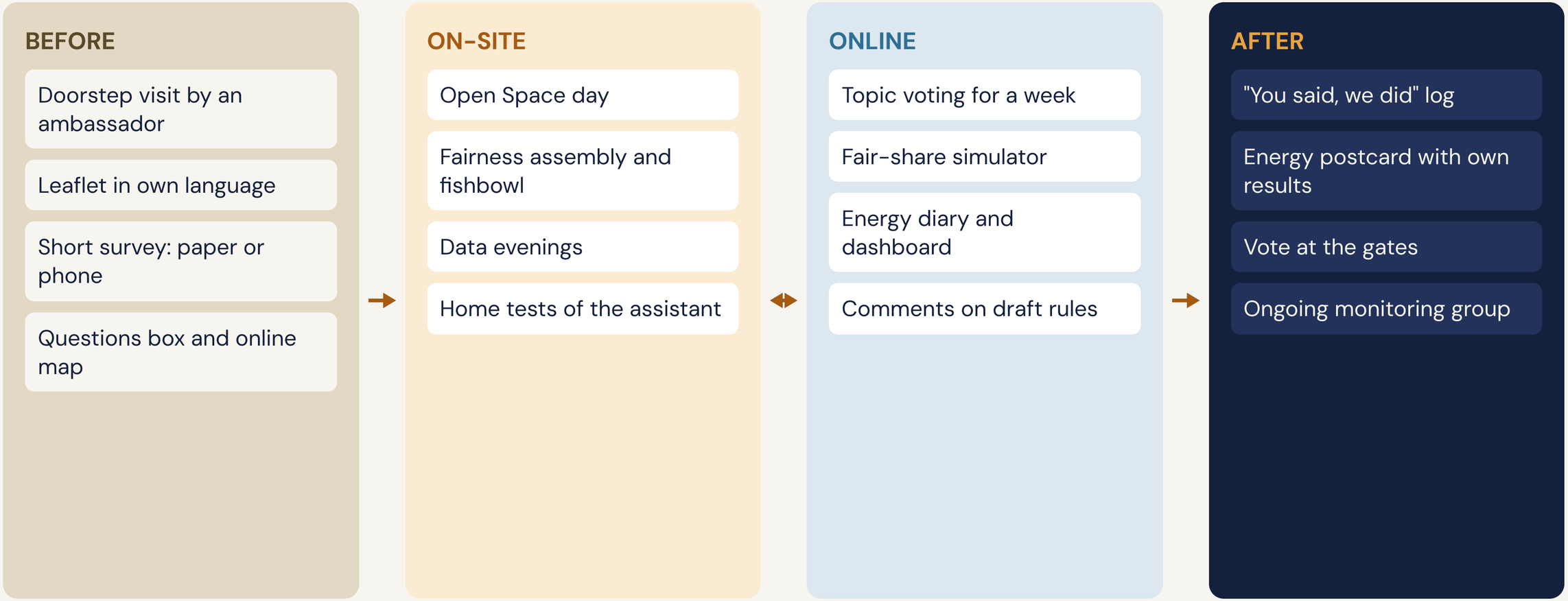
Proposals, comments, simulator and data dashboard

Phone line

A person answers set hours; can vote and report for you

DELIVERABLE E · HYBRID JOURNEY MAP

One tenant's path through the process



Bridge between on-site and online: ambassadors, the bulletin and a 48-hour rule for posting results both ways.

ONLINE ACCESS IS NOT INCLUSION

Alternatives for low digital skills

Assistant without an app

Weekly printed "sunny hours" calendar, SMS or a phone call instead of notifications.

Simulator on paper

Printable household cards and coins to play the fairness rules at the table.

Digital help desk

Ambassadors help on request, building digital skills instead of replacing them.

Equal weight

A paper vote counts exactly like an online one, and is counted in public.

07

Citizen science component

“Energy Detectives”: tenants produce the knowledge the AI is missing

RESEARCH QUESTIONS

Three things we need to know

RQ1 When do households really use energy, and what stops them from using it in sunny hours?

RQ2 Where does the AI forecast go wrong, and for which kinds of households?

RQ3 How would each fairness rule change what real households receive?

THE SIX-PART DESIGN

From question to change

Part	Our answer
Question	RQ1–RQ3: routines, forecast errors, effects of fairness rules
Protocol	Co-written in a workshop by tenants and researchers; tested with 5 households first
Participation	Postcards for all; smart plugs, diaries and shade photos for volunteers
Validation	Diaries checked against meter data; training; uncertainty shown openly
Interpretation	Monthly data evenings where tenants and researchers read results together
Feedback	Findings change the model, the messages and the fairness rule, logged publicly

WHAT CITIZENS ACTUALLY DO

Observe, measure, document, classify, interpret

WIDE TRACK · ALL HOUSEHOLDS

Document: a two-week energy postcard, ticking when big appliances run.

Classify: "could I have moved this to midday? yes / no / why".

Consent: optional sharing of 15-minute meter data.

DEEP TRACK · ~30 VOLUNTEERS

Measure: smart plugs on the washing machine or water heater.

Observe: photos of roof shade at fixed times.

Interpret: label data, compare it with the AI forecast, propose changes.

DATA QUALITY AND UNCERTAINTY

Credible enough for ATER and GSE to listen

Who defines the protocol

Tenants and researchers together, then a 5-household pilot to fix unclear steps.

Cross-checks

Postcards and diaries compared with meter data; mismatches discussed, not deleted.

Training and support

A 30-minute session, a picture guide and an ambassador to call.

Honest uncertainty

Every chart shows how sure we are; small samples are labelled as such.

AVOIDING THE EXTRACTIVE MODEL

Tenants are not unpaid sensors

Data belongs to the CER

The provider only processes it, under the tenants' data charter.

Privacy by design

Aggregated by default, deleted after the pilot, GDPR consent per use.

Opt out, no penalty

Not sharing data never reduces a household's share.

Results returned

Each household gets its own "energy postcard" in plain language.

Right to challenge

Any finding can be contested at a data evening and re-checked.

Recognition

Named credit, certificates and a small allowance for volunteers.

HOW KNOWLEDGE ENTERS DECISIONS

From a postcard to a change in the AI

1 · DATA

Postcards, plugs and photos collected

→

2 · DATA EVENING

Tenants and researchers interpret together

→

3 · PROPOSAL

A written change request with evidence

→

4 · DECISION

Assembly or provider must answer within a month

→

5 · CHANGE LOG

Public record: accepted, changed, or refused with reasons

EXAMPLE

Diaries show shift workers can't shift laundry to noon → the assembly adds a "flexibility credit" for night-time water-heater timers → the AI learns a new household type.

08

Prototype: the Fair Share Simulator

Tenants try fairness rules on real-looking families before they vote

WHAT IT IS

Make the fairness choice tangible

The problem it solves

"Fair sharing" is abstract. Tenants can't vote on a formula they can't see working.

The simulator shows, family by family, what each rule would mean in euros.

How it works

1. Pick a sunny or cloudy month.

2. Mix the fairness rules with sliders.

3. See each household's share change live, and why.

THE RULES TENANTS CAN MIX

Four ideas of what is fair

Equal

Every household gets the same amount.

Simple, but ignores need and effort.

Need

More for low-income or larger households.

Fights energy poverty, needs sensitive data.

Effort

More for those who use energy in sunny hours.

Raises the bonus, penalises shift workers.

Floor

A minimum guaranteed to everyone, whatever the mix.

Protects those who can't shift.

Tenants set the weight of each rule; the simulator shows the trade-offs. The final mix is decided at Gate 2.

FEATURES

Built to be explained, not just used

Household cards

Six personas: a pensioner, a shift worker, a large family and more

Sun and demand view

A day chart: solar production against each family's use

Rule mixer

Sliders for equal, need, effort and the floor

"Why this amount?"

Each share broken down in plain words

Privacy switch

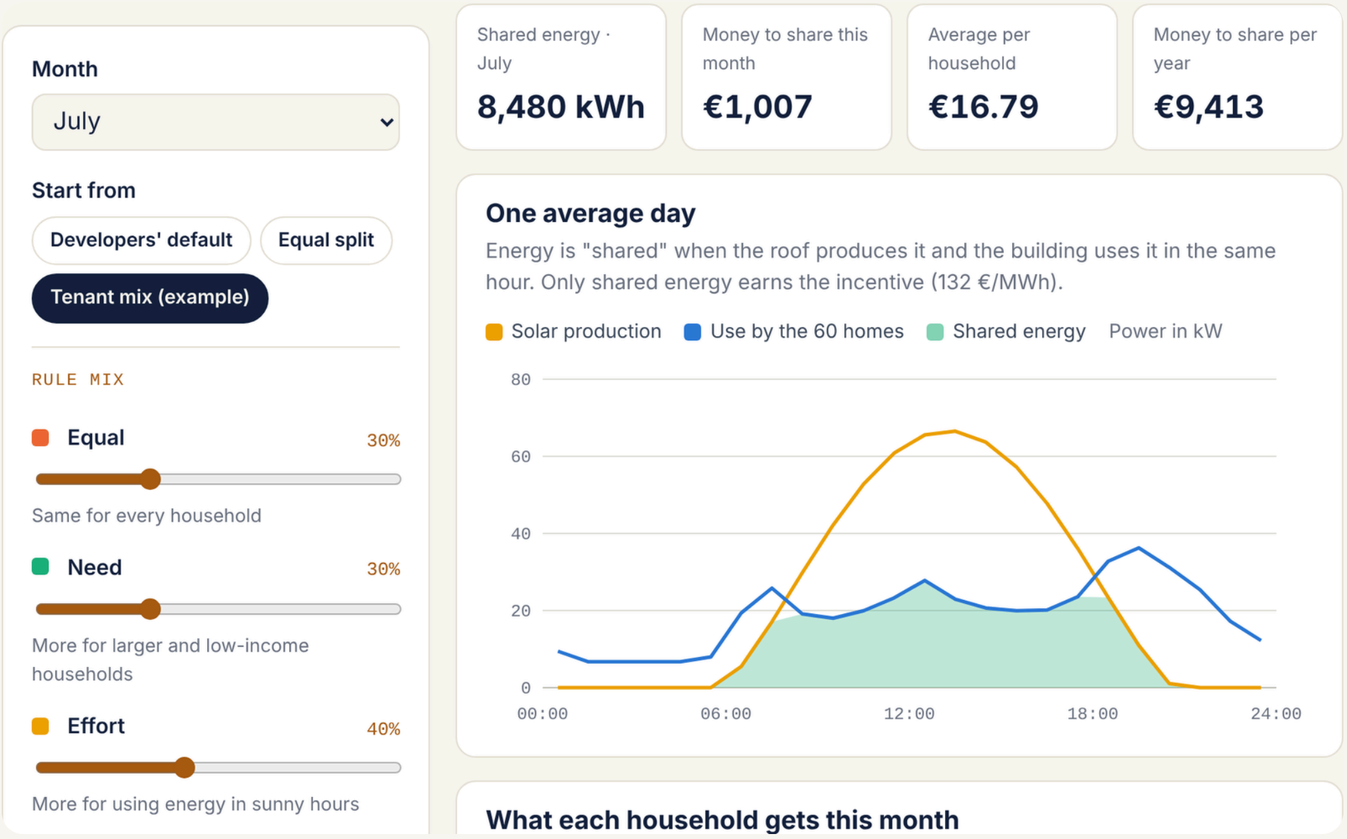
Shows what data each rule needs to work

Paper mode

Printable cards and tables for the table version

DELIVERABLE F · WORKING PROTOTYPE

Try it: the Fair Share Simulator



1 · Pick a month

See solar production, household use and the shared energy that earns the incentive.

2 · Mix the rules

Equal, Need, Effort and a Floor, or a preset: the developers' default or a tenant mix.

3 · Ask why

Every amount is explained in plain words, with the data each rule needs.

kuhyar-saeedi.github.io /next-wave/simulator.html

HOW WE WOULD TEST IT

Three tests with real participants

Test	What participants do	What we measure
Understanding	Explain in their own words why a family got its amount	Share of correct explanations
Usability	Elderly and non-digital tenants use paper or screen with a facilitator	Tasks completed, help needed
Deliberation	The fairness assembly uses it to compare two rules	Did views change? Was the vote more informed?

Before the real process, we can run a first test with classmates playing the household personas.

WHAT TESTING TEACHES US

Questions the prototype answers

Which rule do tenants prefer?

And does the preference change once they see the effect on others?

Is the algorithm explainable?

If tenants can't explain it, the final AI must be simpler.

What data are tenants willing to share?

The answer shapes the data charter (Gate 3).

Does paper work as well as screen?

If not, the hybrid design needs more support.

09

Facilitation and inclusion plan

Reaching beyond the usual participants, and keeping the room fair

1 · RECRUITMENT

Go to people, don't wait for them

Group	How we reach them
Elderly tenants	Doorstep visits, the parish, the local senior centre, phone line
Migrant families	Cultural mediators, community associations, translated leaflets
Shift workers	Evening and weekend slots, WhatsApp voice notes, async voting
Women carers	School gates, childcare during sessions, daytime meetings
Young people	Paid energy ambassador roles, schools, sports clubs
Irregular tenants	Trusted third-sector groups; no ID needed to take part in discussion

Target: the fairness assembly mirrors the complex by age, household type, origin and building.

2 · ACCESSIBILITY

Remove the barriers before people meet them

Language

Plain Italian, key materials translated, mediators at sessions

Timing

Each activity offered twice: weekday evening and Saturday morning

Mobility

Ground-floor rooms in the complex; home visits if needed

Disability

Large print, audio versions, accessible routes and seating

Digital access

Every step works on paper, phone or in person (Step 6)

Care and cost

Childcare, snacks, and a small voucher for long sessions

3 · POWER BALANCE

Experts on tap, not on top

Neutral facilitators

Not from ATER or the provider; trained in conflict and inclusion.

Tenants speak first

Experts answer questions; they don't open discussions.

No-jargon rule

Anyone can raise a "say it simply" card at any time.

Tenants hold the vote

On fairness and data, only tenants vote; institutions give opinions.

4 · CONFLICT

Disagreement is recorded, not hidden

1 · NAME IT
The facilitator writes each disagreement in a public log

→

2 · EXPLORE IT
Fishbowl: both sides explain their reasons

→

3 · TEST IT
Simulator or pilot data where facts can help

→

4 · DECIDE IT
Vote at the gate; minority views attached to the decision

IF TENANTS DISAGREE WITH THE DEVELOPERS

Gates are binding: the provider adapts, or the pilot is paused. An external mediator (e.g. the university) is called before any override.

5 · TRACEABILITY · 6 · FEEDBACK

Every contribution can be followed

Traceability

Each proposal gets a number, on paper and online.

Its status is public: received, discussed, adopted, changed or refused, with reasons.

Feedback

Within 48 hours: session results posted on both channels.

Monthly: "you said, we did" bulletin and a courtyard return event.

End of pilot: each household's energy postcard.

10

Evaluation framework

What changed in the technology because people took part

SIX DIMENSIONS

What we measure, and how

Dimension	Indicator	How we measure it
Inclusion	Participants vs the complex by age, origin, household, building	Sign-in sheets, assembly lottery records
Dialogue	Share of speaking time by tenants; disagreements logged and answered	Facilitator logs, disagreement log
Knowledge	Citizen data used in model, messages or rule decisions	Change log references to citizen findings
Learning	Understanding of the CER and the AI, before vs after	Short quiz and interviews, start and end
Innovation	Features, rules or governance changed by participation	Before/after comparison of the design
Continuity	Active monitoring group and ambassadors after the pilot	Attendance 3 and 6 months later

THE BEFORE / DURING / AFTER TEST

Documenting the trace

BEFORE

Freeze the developers' starting design: incentive-maximising AI, equal split, full automation, app only, provider holds the data.

Signed and dated.

DURING

Record every proposal, data finding, controversy, prototype test and rejected alternative in the change log.

AFTER

Compare the final design with the frozen one, line by line, and trace each change back to a contribution.

WHAT PARTICIPATION COULD CHANGE

The expected trace in the technology

Element	Before (developers)	Possible after (participation)
Fairness	Equal split	Mixed rule with a minimum floor for all
Control	Automates appliances	Advises by default; automation opt-in
Access	App only	App, SMS, paper calendar, phone line
Data	Provider holds all data	CER owns data under a tenant charter
Model	Generic household profiles	New types learned from diaries (e.g. shift work)
Governance	Board of ATER and provider	Elected tenant seats and an appeals route

WHAT WOULD COUNT AS EVIDENCE

Participation worked if…

- At least one design element changed and can be traced to tenants' input
- The fairness assembly mirrored the complex, not only the usual participants
- Vulnerable households' share was not lower than under the developers' default
- Tenants can explain the rule that shares their money
- A tenant group keeps monitoring after the pilot ends

THE UNRESOLVED DILEMMA

Fairness versus a bigger bonus

The tension

Rewarding effort raises the community's total income. Protecting those who can't shift lowers it. Tenants may choose a fairer but smaller pie.

The legal limit

Tenants without a regular contract may not be able to join the CER. Participation can include them in the discussion, but not, alone, in the benefits.

We accept both as open questions, and our process makes them visible instead of hiding them in code.

THE ANSWER TO THE COURSE QUESTION

What will be different because people participated?

The rule

The money would be shared by a fairness rule tenants chose and voted on, not one written by the provider.

The data

The AI would learn from tenant-collected knowledge about real daily life, under data rules tenants agreed.

The access

Elders, shift workers and non-Italian speakers could use and control the assistant without a smartphone.

Participation did not decorate the technology. It is designed to change it.